

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) on a provider-facing clinical decision support system, owning problems end-to-end from API to UI in a fast-moving, ambiguous environment.
- Extending a **Firebase Cloud Functions** backend where the LLM (**Gemini**) only synthesizes while the backend deterministically controls all eligibility and coverage logic, keeping a safety-sensitive system verifiable rather than a black box.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (Python, **FastAPI**, sentence-transformers, React) end to end, owning architecture and all technical decisions delegated by the PI.
- Built the platform's engineering backbone: a YAML model registry as single source of truth, a versioned construct library (**99 constructs from 38 questionnaires**), and structured data-quality warnings, growing the backend to **40 hermetic CI tests** with no ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack internal tooling platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.

Projects

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication from a naive baseline through cache-aware tiling, manual **AVX/SIMD** vectorization with register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB misses with **Linux perf** and inspecting generated assembly to close the gap toward peak throughput.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Cut end-to-end Spark ETL runtime **~25%** (17–19 min to 14.3 min) on a **~1.4B-row** dataset via broadcast joins (shuffle minutes to ~0.05s), AQE, skew-join handling, and partition/column pruning, using Spark UI stage metrics to localize bottlenecks and document a repeatable optimization playbook.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a cross-platform desktop app running a fully offline on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local TTS) with a FastAPI backend and async queues that keep the UI responsive under real-time video and audio, with no external network dependencies.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: C++, Python, TypeScript, JavaScript, Java, SQL, Bash, PostgreSQL, MongoDB, SQLite

Systems & Perf: Linux perf, AVX/SIMD intrinsics, OpenMP, multithreading, Spark / PySpark, cache-aware optimization, async pipelines

Backend & APIs: FastAPI, Node.js, Express, REST APIs, async services, integration testing, Next.js, Docker

ML & AI: PyTorch, NumPy, sentence-transformers, LLM agents, RAG, evaluation, provenance

Cloud & DevOps: Git, GitHub Actions, CI/CD, Docker, Kubernetes, AWS, GCP, Azure, Linux

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.